

TASK 2

Chapter 11 - Database Performance Tuning and Query Optimization

Instructions:

Document layout and submission due date must strictly adhere to the following rules:

Document Presentation:

- Your student number, name, and signature must be on the top right-hand side of Page 1.
- The task must be handwritten using a black pen (typed work will not be accepted)
- Write neatly and leave a line to separate questions.

Document Submission Due Date:

- One handwritten, scanned PDF Soft copy must be submitted on Moodle
 - The PDF soft copy must be saved as your student number followed by Task2 (e.g. **26658193_Task2**)
 - Ensure that you are uploading the correct document to Moodle
 - The task must be submitted on Moodle by Thursday, 27 August 2026. Moodle closes at 15:00.
 - Once Moodle closes, no reopening; no emailed submissions. Emailed submissions will be deleted without being opened
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Answer the following questions:

- 1 . List and describe some typical DBMS processes.
2. Describe query optimization and the modes that an optimizer can operate in.
3. What is the difference between SQL performance tuning and database performance tuning activities
4. What is the focus of most performance tuning activities, and why does that focus exist?
5. The DBMS processes queries in three phases. What are those phases, and what is accomplished in each phase?
6. How are database statistics obtained?
7. What are database statistics, and why are they important?
8. What database statistics measurements are typical of tables, indexes, and resources?
9. What are some general guidelines for creating and using indexes?
10. If indexes are so important, why not index every column in every table? (Include a brief discussion of the role played by data sparsity.)
11. What is the difference between a rule-based optimizer and a cost-based optimizer?
12. What are optimizer hints and how are they used?
13. What does RAID stand for and what are some commonly used RAID levels?